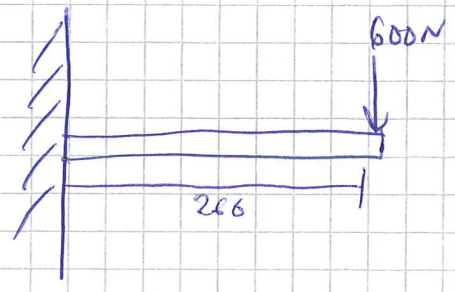
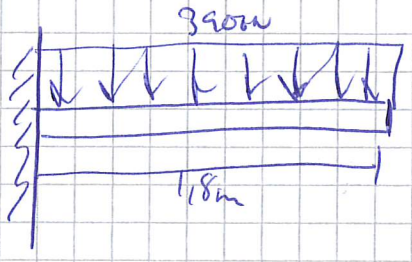
48



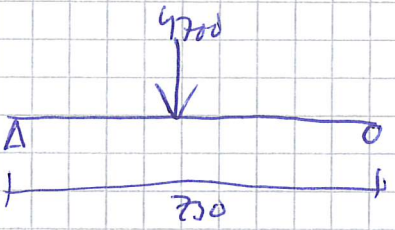
$M_b = F \cdot L = 600 \cdot 260 = 158000\ Nm$

49



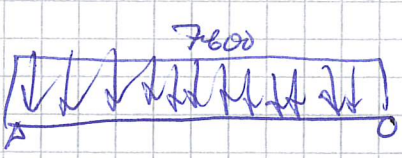
$M_b = \frac{QL}{2} = \frac{3600 \cdot 1,8}{2} = 3570\ Nm$

50



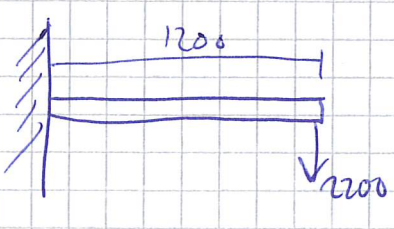
$M_b = \frac{FL}{4} = \frac{4700 \cdot 0,73}{4} = -857\ Nm$

51



$M_b = \frac{QL}{8} = \frac{7600 \cdot 0,97}{8} = 921,5\ Nm$

52



$R_e = 240\ N/mm^2$

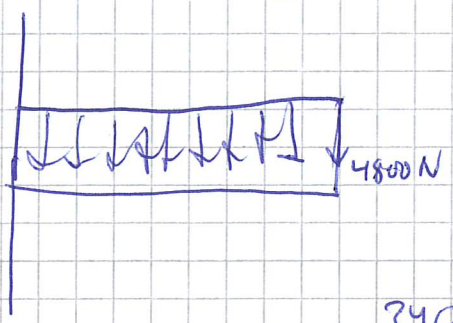
$n_s = 5$

$B = 20$ Solut H

$\frac{R_e}{n_s} = \frac{M_b}{W_b}$ $\quad \frac{240}{5} = \frac{2200 \cdot 1200}{BH^2/6} \Rightarrow 48 = \frac{1,584 \cdot 10^7}{BH^2}$

$\Rightarrow \frac{1,584 \cdot 10^7}{48} = BH^2 \Rightarrow H = \sqrt{16500} = 128mm$

53 $L = 1400\,mm$



$R_e = 340\,N/mm^2$ $n_s = 1,8$

sökt D

$$\frac{340}{1,8} = \frac{M_b}{W_b} = 188,89$$

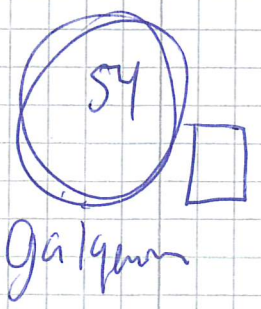
$$M_b = \frac{QL}{2} \qquad W_b = \frac{\pi D^3}{32}$$

$$M_b = \frac{4800 \cdot 1400}{2} = 3,36 \cdot 10^6\,Nmm$$

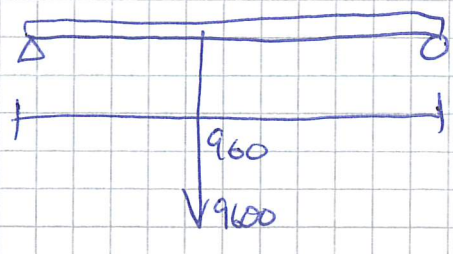
$$188,89 = \frac{3,36 \cdot 10^6 \cdot 32}{\pi D^3} \Rightarrow 188,89\,\pi D^3 = 3,36 \cdot 10^6 \cdot 32 \Rightarrow$$

$$\Rightarrow D = \sqrt[3]{\frac{3,36 \cdot 10^6 \cdot 32}{188,89\pi}} = 56,6\,mm \Rightarrow 57\,mm$$

54



galgenn



$h = 50\,mm$

$B = ?$

$R_c = 420\,N/mm^2$

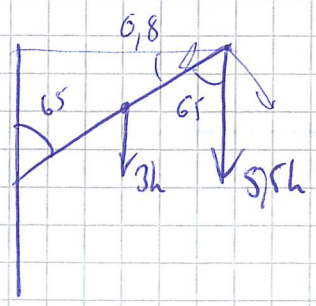
$n_s = 2$

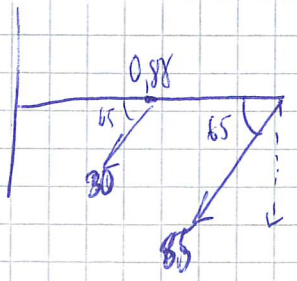
$$\frac{R_c}{n_s} = \frac{M_b}{W_b} \Rightarrow \frac{M_b \cdot n_s}{R_c} = W_b \qquad W_b = \frac{B H^2}{6}$$

$$\frac{M_b \cdot n_s}{R_c} = \frac{B H^2}{6} \Rightarrow \frac{6 M_b n_s}{R_c \cdot B} = H^2 \Rightarrow$$

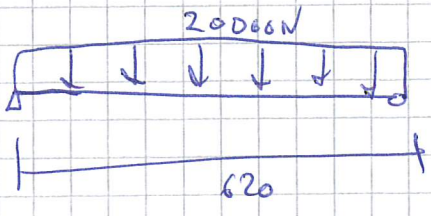
$$B = \frac{M_b \cdot n_s \cdot 6}{R_c \cdot H^2} = B \Rightarrow \frac{\frac{9600 \cdot 960}{4} \cdot 2 \cdot 6}{420 \cdot 50^2} = 26{,}33$$

$= 26{,}4\,mm$





55



$D = 60\,mm$

$d = ?$

$R_e = 360\ N/mm^2$

$n_s = 1{,}4$

$$\frac{R_e}{n_s} = \frac{M_b}{W_b}$$

$$W_b = \frac{M_b \cdot n_s}{R_e} \qquad \frac{\pi(D^4 - d^4)}{32D} = \frac{M_b n_s}{R_e} \Rightarrow$$

$$\Rightarrow \frac{M_b \cdot n_s \cdot 32D}{R_e \cdot \pi} = D^4 - d^4 \Rightarrow$$

$$d = \sqrt[4]{\frac{M_b \cdot n_s \cdot 32D}{R_e \cdot \pi} + D^4} = 55{,}2$$

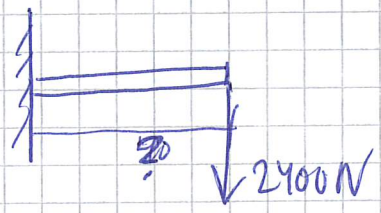
$$\frac{60 - 55{,}2}{2} = 2{,}4\,mm$$

56

$D = 40\,mm$

$R_e = 380\ N/mm^2$

$n_s = 1{,}8$



$M_b = F \cdot l$

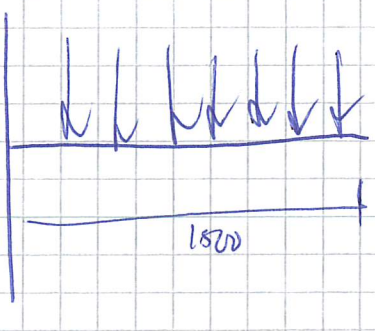
$W_b = \frac{\pi D^3}{32}$

$$\frac{R_{es}}{n_s} = \frac{M_b}{W_b}$$

$$211 = \frac{2400L \cdot 32}{\pi 40^3} \Rightarrow \frac{211 \cdot \pi \cdot 40^3}{32 \cdot 2400} = L \Rightarrow$$

$$L = 552{,}4\,mm$$

57



$M_b = \frac{QL}{2}$

$R_e = 440\ N/mm^2$

$L = 1500\ mm$

$n_s = 3$

Soluț (F)

$D = 80$

$d = 65$

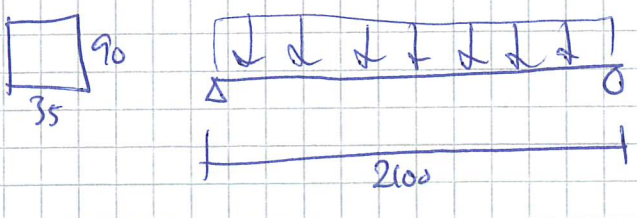
$\frac{R_e}{n_s} = \frac{M_b}{W_b}$

$\frac{440}{3} = 146{,}67 = \frac{Q \cdot 1500}{2 \cdot 2 \cdot \pi \left(\frac{80^4 - 65^4}{32 \cdot 80}\right)}$ $\Rightarrow Q = 11092\ N =$

113437

$= 1{,}1\ kN$

58.



$R_e = 280\ N/mm^2$

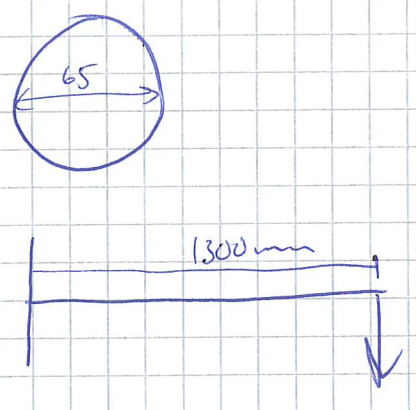
$n_s = 2,2$

$$\frac{R_e}{n_s} = \frac{M_b}{W_b} \Rightarrow M_b = \frac{R_e \cdot W_b}{n_s} \Rightarrow \frac{QL}{8} = \frac{R_e \cdot W_b}{n_s} \Rightarrow$$

$$\Rightarrow Q = \frac{R_e \cdot W_b \cdot 8}{n_s \cdot L} = \frac{280 \cdot \frac{35 \cdot 90^2}{6} \cdot 8}{2,2 \cdot 2100} =$$

$= 22909\ N = 2291\ kg \Rightarrow 2,3\ ton$

59.

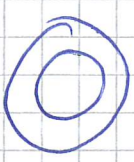


$R_e = 380\ N/mm^2$

$n_s = 1,8$

$$\frac{380}{1,8} = \frac{1300 \cdot F \cdot 32}{\pi \cdot 65^3} \Rightarrow F = \frac{380 \cdot \pi \cdot 65^3}{1,8 \cdot 1300 \cdot 32} = 4378\ N$$

60

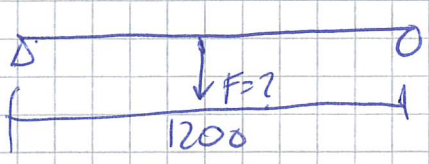


$D = 68\,mm$

$d = 56\,mm$

$R_e = 320\,N/mm^2$

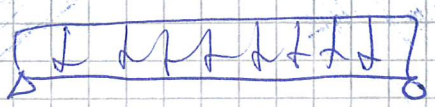
$n_s = 2{,}5$



$$\frac{320}{2{,}5} = \frac{320 \cdot F \cdot 1200}{4 \cdot \pi (D^4 - d^4)} \Rightarrow F = \frac{320 \cdot 4\pi (68^4 - 56^4)}{2{,}5 \cdot 320 \cdot 1200} = \underline{\underline{712\,N}}$$

$= 710\,kg$

61



$L = 1400\,mm$

$d = 45\,mm$

$R_e = 280\,N/mm^2$

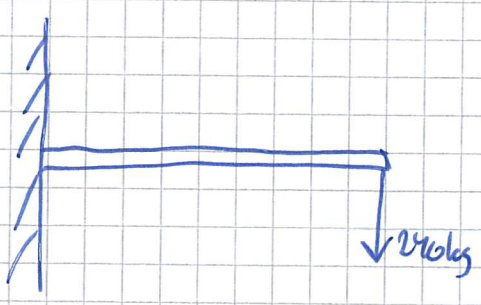
$n_s = 1{,}6$

Sökt Q

$$\frac{R_e}{n_s} = \frac{QL \cdot 32}{8\pi D^3} \Rightarrow 175 = \frac{QL32}{8\pi D^3} \Rightarrow$$

$$\Rightarrow Q = \frac{175 \cdot 8 \cdot \pi \cdot D^3}{32L} = \frac{175 \cdot 8 \cdot \pi \cdot 45^3}{32 \cdot 1400} = \underline{\underline{8946\,N}} = 895\,kg$$

62.



D = 74mm

d = 66mm

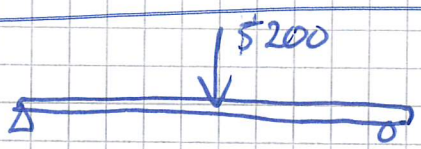
L = 800mm

Sökt R_e

$$R_e = \frac{M_b}{W_b} = \frac{F \cdot L}{\frac{\pi(D^4 - d^4)}{32D}} \Rightarrow$$

$$\Rightarrow \frac{2400 \cdot 800 \cdot 32 \cdot 74}{\pi(74^4 - 66^4)} = 131 \text{ N/mm}^2$$

63.



d = 68mm

L = 1300mm

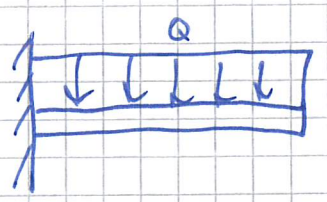
$n_s = 1{,}6$

Sökt R_e

$$\frac{R_e}{n_s} = \frac{M_b}{W_b} \Rightarrow R_e = \frac{M_b \cdot n_s}{W_b} = \frac{5200 \cdot 1300 \cdot 1{,}6 \cdot 32}{4 \cdot \pi \cdot 68^3} =$$

$$= 87{,}6 \text{ N/mm}^2$$

64.



$L = 2000$

$b = 50$

$h = 90$

$n_s = 2,2$

$Q = 12000\ N$

$R_c = ?$

$$\frac{R_c}{n_s} = \frac{M_b}{W_b} \Rightarrow R_c = \frac{M_b \cdot n_s}{W_b} = \frac{12000 \cdot 2000 \cdot 2,2 \cdot 6}{2 \cdot 50 \cdot 90^2} =$$

$$= 391\ N/mm^2$$

65.

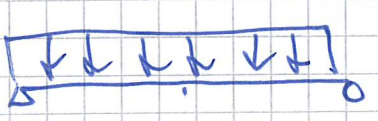
$R_c = ?$

$d = 70mm$

$L = 1500mm$

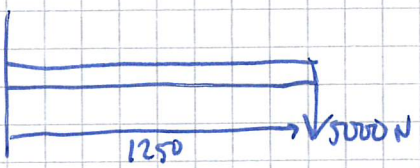
$n_s = 1,8$

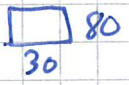
$F = 18000N$



$$\frac{R_c}{n_s} = \frac{M_b}{W_b} \Rightarrow R_c = \frac{18000 \cdot 1500 \cdot 32 \cdot 1,8}{8 \cdot \pi \cdot 70^3}$$

66.





$n_s = ?$ $\quad R_c = 320\ N/mm^2$

$$\frac{320}{n_s} = \frac{5000 \cdot 1250 \cdot 6}{30 \cdot 80^2} \Rightarrow \frac{320 \cdot 30 \cdot 80^2}{5000 \cdot 1250 \cdot 6} = 1,63$$